

APM02B2 Introduction to Numerical Analysis

Fixed Point Iteration

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It is better to do the right problem
the wrong way than the wrong
problem the right way.

Richard W. Hamming

Problem Statement

Problem Statement

Let $I \subset \mathbb{R}$ and $f : I \rightarrow \mathbb{R}$ be continuous on I .

Generally, we assume that I is a bounded and closed interval. That is,

$$I = [a, b],$$

where $a, b \in \mathbb{R}$ such that $a < b$ (this guarantees that I is not empty).

Problem

Consider the general equation

$$f(x) = 0. \tag{1}$$

We wish to find all $x^* \in I$ that satisfy equation (1). These x^* are called the *roots of the function f* .

Fixed Point Theory

Fixed Points

An equivalent manner to solve (1) is by rewriting it in the form

$$x = g(x), \tag{2}$$

where $g : [a, b] \rightarrow \mathbb{R}$ is a continuous function on $[a, b]$ satisfying certain properties (which we shall discuss).

Equation (2) is referred to as a *fixed-point equation*.

Definition (Fixed point of a function)

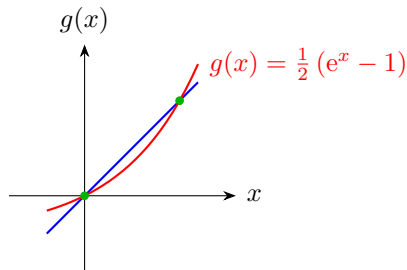
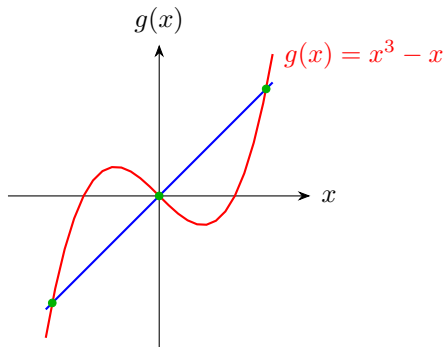
The solution x^* to equation (2) is called a *fixed point of g* . That is, it is a number x^* satisfying $x^* = g(x^*)$.

A fixed point for a function is a number at which the value of the function does not change when the function is applied.

Fixed Points

Visualisation

We can visualise **fixed points** as the intersections of the graphs $y = g(x)$ and $y = x$.



Fixed Points

Finding fixed points

Example

Find the fixed point(s) of the function $g(x) = x^3 - x$.

Substituting the given function into the fixed-point equation (2) yields the equation

$$x = x^3 - x,$$

which we solve for x :

$$x = x^3 - x \quad \implies \quad 0 = x^3 - 2x = x(x^2 - 2) \quad \implies \quad x^* = 0, \quad x^* = \pm\sqrt{2}.$$

Fixed Points

Finding fixed points

If g is a polynomial of degree < 5 , we could find the fixed points algebraically via factorisation or long division.

However, if g is a polynomial of degree ≥ 5 , or a nonlinear function, we would need to resort to approximating the fixed points (see slide 18).

A lot can still be said about the fixed points themselves without solving equation (2) by considering the properties of the function g .

Fixed Points

Existence and uniqueness

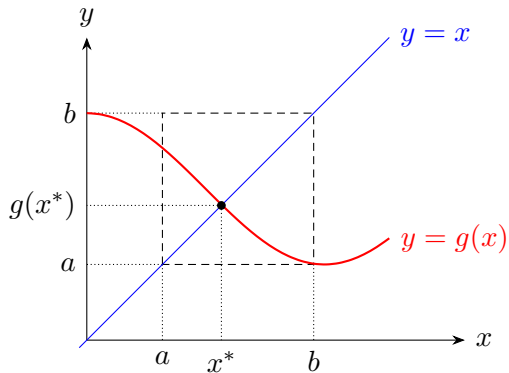
A function may have none, one, or many fixed points.

The following questions now arise:

- Under which circumstances do fixed points exist for a given function g ?
- What conditions must g satisfy for a fixed point to be unique?

Fixed Points

Existence and uniqueness



Suppose $g : [a, b] \rightarrow \mathbb{R}$ is continuous on $[a, b]$.

If $g(a) = a$ or $g(b) = b$, the g has a fixed point at the end point of the interval $[a, b]$.

Fixed Points

Existence and uniqueness

If this is not the case, then suppose $g(a) > a$ and $g(b) < b$.

Define $h(x) = g(x) - x$. This function is continuous on $[a, b]$. (Why?) It also satisfies $h(a) = h(a) - a > 0$ and $h(b) = h(b) - b < 0$.

Since $h(a)h(b) < 0$, the intermediate value theorem implies that f has a root $x^* \in (a, b)$ for which $h(x^*) = 0$.

This number x^* is a fixed point of g , since

$$0 = h(x^*) = g(x^*) - x^* \quad \implies \quad x^* = g(x^*).$$

Fixed Points

Existence and uniqueness

We have proved the following

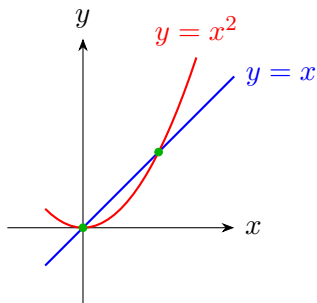
Theorem (Existence of a fixed point)

Let $g : [a, b] \rightarrow \mathbb{R}$ be continuous on $[a, b]$ and $g(x) \in [a, b]$ for all $x \in [a, b]$. Then g has at least one fixed point in $[a, b]$.

This theorem gives the condition under which a fixed point exists.

Fixed Points

Existence and uniqueness



Now consider, for example,
 $g(x) = x^2$ on $[0, 1]$.

We can easily verify that
 $g(x) \in [0, 1]$ for all $x \in [0, 1]$.
However, as can be seen from the
figure on the left, g has *two* fixed
points in $[0, 1]$.

To solve equation (1) via equation
(2), we would rather have a *unique*
solution^a and therefore a unique
fixed point.

^aTo see why we want a unique solution,
see slides [insert slide numbers](#).

Fixed Points

Existence and uniqueness

Let us now determine the circumstances under which a fixed point of g would be unique.

Fixed Points

Existence and uniqueness

We have proved the following

Theorem (Uniqueness of a fixed point)

Let $g : [a, b] \rightarrow \mathbb{R}$ be continuous on $[a, b]$ and $g(x) \in [a, b]$ for all $x \in [a, b]$. If g' exists on (a, b) and there exists a constant $L \in (0, 1)$ such that

$$|g'(x)| \leq L, \text{ for all } x \in (a, b),$$

then the fixed point x^ of g in $[a, b]$ is unique.*

Fixed Point Iteration

Fixed Point Iteration

Equation (2) suggests the following iterative method:

$$x_{k+1} = g(x_k) \quad (3)$$

We refer this numerical method as a *fixed-point iteration*.

Equation (3) is also called a *simple iteration* and the numbers x_k of the sequence it generates are called *iterates*¹.

¹Simple iteration and iterates are more generally used for iterative methods and the elements of the sequences these method generate.

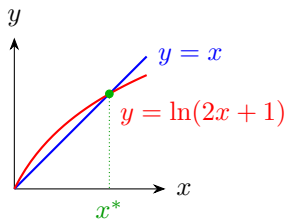
Fixed Point Iteration

Example

Find a root of $f(x) = e^x - 2x - 1$ in $[1, 2]$ using fixed-point iteration with initial value $x_0 = 1$.

Rewrite $f(x) = e^x - 2x - 1 = 0$ as

$$e^x = 2x + 1 \quad \implies \quad x = \ln(2x + 1) \quad \implies \quad g(x) = \ln(2x + 1)$$



Fixed Point Iteration

Example (continued)

Using the initial value $x_0 = 1.0$, we obtain for the first five iterations

$$x_1 = g(x_0) \approx 1.0986, \quad |x_1 - x_0| \approx 0.0986$$

$$x_2 = g(x_1) \approx 1.1623, \quad |x_2 - x_1| \approx 0.0637$$

$$x_3 = g(x_2) \approx 1.2014, \quad |x_3 - x_2| \approx 0.0391$$

$$x_4 = g(x_3) \approx 1.2381, \quad |x_4 - x_3| \approx 0.0232$$

$$x_5 = g(x_4) \approx 1.2460, \quad |x_5 - x_4| \approx 0.0136$$

Since a tolerance was not specified in the problem statement, we can either stop here or continue with the iteration until we meet our own specified tolerance.

To calculate the root to within a tolerance of 10^{-15} (the most accurate we can do on an average laptop) will take 58 iterations.

Fixed Point Iteration

Potential difficulty

One of the difficulties with fixed-point iteration is that there are potentially multiple ways of constructing g from f .

The following is taken from Atkinson 1993, p.83. Consider solving the equation $x^2 - 5$ for the root $x^* = \sqrt{5}$. There are four possible fixed-point iteration formulae that may be used to solve this equation:

$$\begin{aligned}x_{k+1} = g_1(x_k) &= 5 + x_k - x_k^2, & x_{k+1} = g_2(x_k) &= \frac{5}{x_k}, \\x_{k+1} = g_3(x_k) &= 1 + x_k - \frac{x_k^2}{5}, & x_{k+1} = g_4(x_k) &= \frac{1}{2} \left(x_k + \frac{5}{x_k} \right).\end{aligned}$$

Fixed Point Iteration

Why use fixed-point iteration?

Advantages:

- Fixed-point iteration (3) is easier to calculate and implement² (provided we know what g is).
- Analysis for existence, uniqueness, and convergence behaviour is more simple than other root-finding methods.

Disadvantages:

- Convergence to a solution is not guaranteed³
- Construction of g may prove difficult⁴.

²As compared to the bisection method.

³Although, we can determine for which x -values a problem may converge.

⁴Or potentially have other unwanted side-effects.

Analysis

Fixed Point Iteration

Convergence analysis

The questions we now wish to answer are:

- Under which conditions does a solution for the fixed-point iteration (3) exist?
- If such a solution exists, under which conditions will be unique?

Fixed Point Iteration

Convergence analysis

To answer the first question, it suffices to show that

$$\lim_{k \rightarrow \infty} x_k = x^*, \quad (4)$$

where $x_k = g(x_{k-1})$.

To prove (4), we consider the absolute error $|x_k - x^*|$. Recall that if x^* is a root of f , then it is a fixed point of g . Now rewrite $|x_k - x^*|$ as

$$|x_k - x^*| = |g(x_{k-1}) - g(x^*)|.$$

Fixed Point Iteration

Convergence analysis

By the mean value theorem, there exists $\xi_{k-1} \in (a, b)$ such that

$$g(x_{k-1}) - g(x^*) = g'(\xi_{k-1})(x_{k-1} - x^*)$$

and, therefore,

$$|x_k - x^*| = |g(x_{k-1}) - g(x^*)| = |g'(\xi_{k-1})| |x_{k-1} - x^*|$$

Fixed Point Iteration

Convergence analysis

If we now assume that $|g'(x)| \leq L$, where $L \in (0, 1)$, for all $x \in (a, b)$, then

$$|x_k - x^*| = |g(x_{k-1}) - g(x^*)| = |g'(\xi_{k-1})| |x_{k-1} - x^*| \leq L |x_{k-1} - x^*|.$$

Hence,

$$|x_k - x^*| \leq L |x_{k-1} - x^*|.$$

Applying this inequality inductively yields

$$|x_k - x^*| \leq L |x_{k-1} - x^*| \leq L^2 |x_{k-2} - x^*| \leq \cdots \leq L^k |x_0 - x^*|$$

Fixed Point Iteration

Convergence analysis

Since $0 < L < 1$, it follows that

$$\lim_{k \rightarrow \infty} L^k = 0$$

and, therefore,

$$\lim_{k \rightarrow \infty} |x_k - x^*| \leq \lim_{k \rightarrow \infty} L^k |x_0 - x^*| = 0.$$

Since $\lim_{k \rightarrow \infty} 0 \leq \lim_{k \rightarrow \infty} |x_k - x^*|$, by the squeeze theorem then follows that

$$\lim_{k \rightarrow \infty} |x_k - x^*| = 0$$

and, therefore,

$$\lim_{k \rightarrow \infty} x_k = x^*.$$

Fixed Point Iteration

Convergence analysis

We have just proved the following

Theorem (Fixed-Point Theorem)

Let $g : [a, b] \rightarrow \mathbb{R}$ be continuous on $[a, b]$ and $g(x) \in [a, b]$ for all $x \in [a, b]$. Suppose that g' exists on (a, b) and $L \in (0, 1)$ exists such that

$$|g'(x)| \leq L, \quad \text{for all } x \in [a, b].$$

Then, for any number $x_0 \in [a, b]$, the sequence defined by

$$x_{k+1} = g(x_k), \quad k \in \mathbb{N},$$

converges to the unique fixed point $x^ \in [a, b]$.*

Questions To Ponder



Exercises

Exercises

Exercises

Further Reading

-  Atkinson, Kendall E. (1993). *Elementary Numerical Analysis*. 2nd ed. John Wiley & Sons.
-  Burden, Richard L. and Douglas J. Faires (2015). *Numerical Analysis*. 10th ed. Cengage Learning.
-  Gerald, Curtis F. and Patrick O. Wheatley (2004). *Applied Numerical Analysis*. Pearson Education, Inc.
-  Hamming, Richard W. (1986). *Numerical Methods for Scientists and Engineers*. Dover Publications, Inc., New York.
-  Isaacson, Eugene and Herbert B. Keller (1994). *Analysis of Numerical Methods*. Dover Publications, Inc., New York.
-  Süli, Endre and David F. Mayers (2003). *An Introduction to Numerical Analysis*. Cambridge University Press.

Fixed Points

Behaviour of fixed points